

Amazon Sales Analysis

Strategic Insights for Revenue & Quality Optimization

Context & Objectives



Business Context

The e-commerce landscape is driven by data. Understanding sales performance, pricing strategies, and customer sentiment is critical for maintaining competitive advantage.

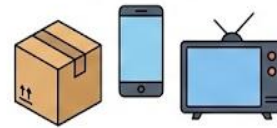
Primary Objectives

- **Sales Performance:** Identify top-performing categories and products.
- **Pricing Strategy:** Analyze the relationship between price, discount, and sales volume.
- **Customer Sentiment:** Understand drivers of negative reviews to improve product quality.

Amazon Sales Data Overview

Key metrics from dataset scope

TOTAL PRODUCTS



1,350+

Unique Products
Analyzed

TOTAL REVIEWS



**~27
MILLION**

Customer Reviews
Processed

TOTAL REVENUE



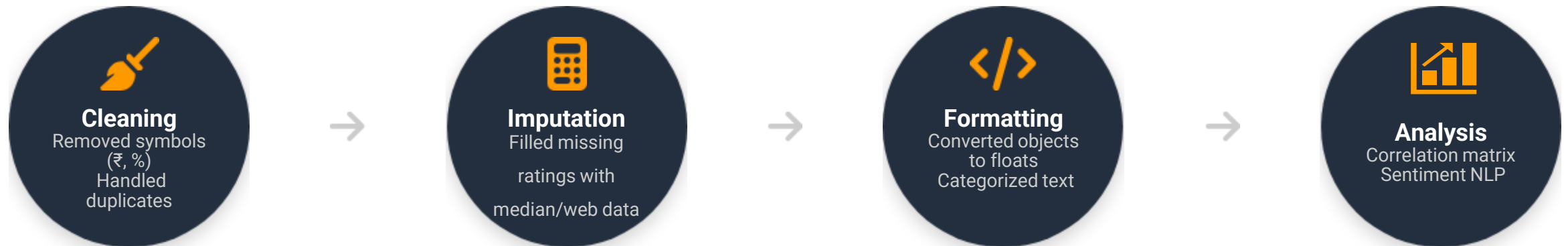
**\$4.58
MILLION**

Sales Revenue
Analyzed

Data Processing Methodology



To ensure analysis accuracy, the raw dataset underwent rigorous cleaning and transformation.



✓ **Result:** A robust, clean dataset ready for high-confidence statistical analysis.

Sales Dashboard Overview



TOTAL REVENUE

\$4.58M

AVG DISCOUNT

47.7%

AVG RATING

4.1 / 5.0

TOTAL PRODUCTS

1.35K

Top Performing Product

Sony Bravia 65" 4K TM

Highest profitability driver in Electronics.

Most Reviewed Product

AmazonBasics HDMI Cable

High volume, essential accessory.

Revenue by Category (in Millions)

Electronics

1.56M

Computers

1.07M

Home & Kitchen

0.65M

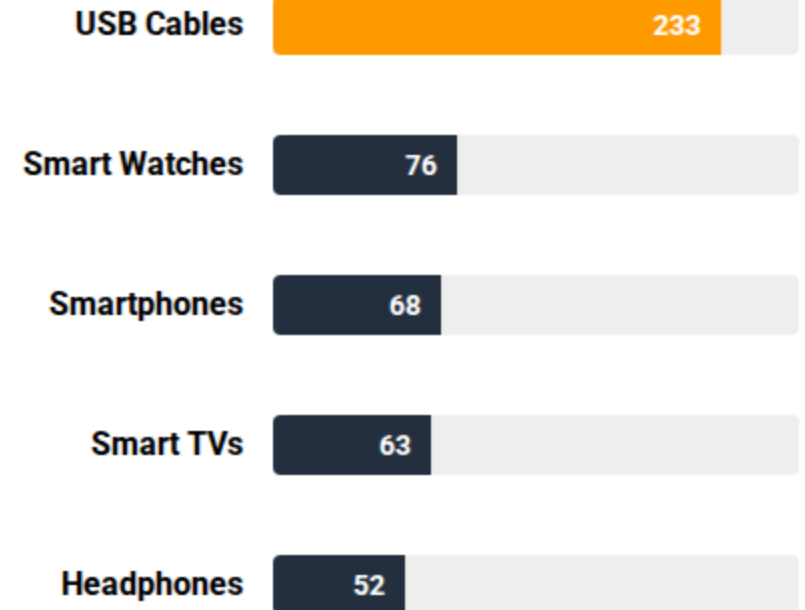
Category Volume Analysis



Electronics and Accessories overwhelmingly dominate the inventory, representing the core of the analyzed sales data. While **Home & Kitchen** has significant product diversity, the volume of reviews and sales velocity is driven by tech consumables like cables and small gadgets.

Insight: Accessories (Cables, Mice) drive volume, while core Electronics drive revenue.

Product Count by Category



Pricing & Discount Correlation



Correlation Matrix

Actual Price 1.0	Discount % -0.118
Discount % -0.118	1.0

Pearson Correlation

The "Premium" Reality

We observed a **weak negative correlation (-0.118)** between Actual Price and Discount Percentage.

Interpretation: Expensive products do *not* rely on heavy discounts to sell. High-value items (TVs, Laptops) maintain value.

Strategic Note: Massive discounts are primarily used to move lower-cost inventory, not premium goods.

Customer Satisfaction Profile



High Overall Satisfaction

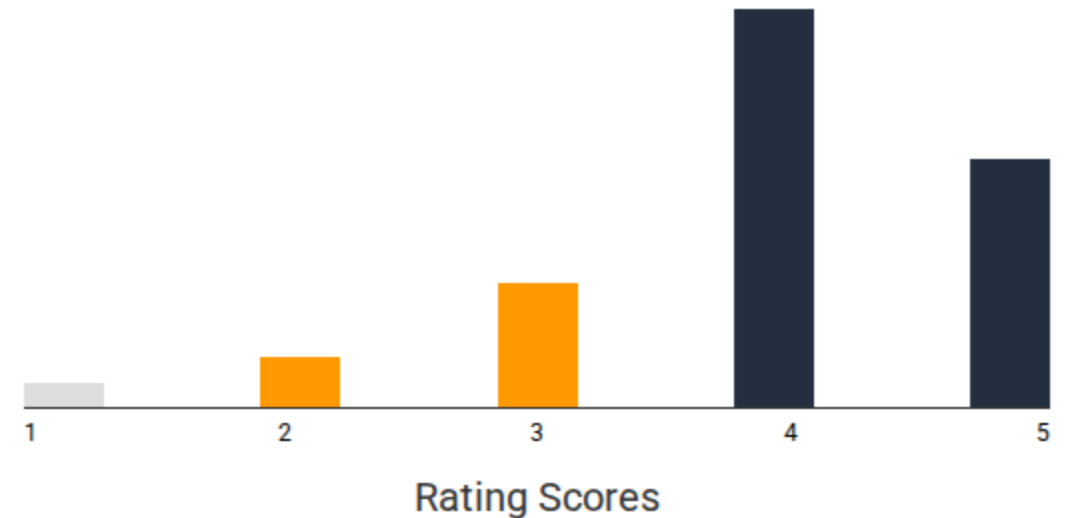
The average product rating is **4.1 out of 5**, indicating a generally healthy product portfolio.

Category Winners:

- Tablets: 4.6★
- Network Adapters: 4.5★
- Storage/Memory: 4.5★

Office products and simple electronics consistently score higher than complex home appliances.

Rating Distribution



Deep Dive: The "Low Rating" Trap



We analyzed products with ratings below 3.0 to understand the drivers of dissatisfaction.

Metric	Low Rated Products (< 3.0)	All Products (Avg)	Variance
Avg Discount %	64%	47.7%	↑+16.3%
Avg Price	\$518	\$3,125	↓83%

⚠ Insight: Heavy discounts on cheap products correlate with poor ratings. Customers perceive these as “cheaply made” or misleading marketing, damaging brand trust.



Using Natural Language Processing (NLP) on review text:

- Conclusion:** Negative sentiment is rarely about shipping or aesthetics; it is overwhelmingly about **product functionality** and **reliability**.

Strategic Recommendations



Quality Control

Implement stricter QC for items priced under \$600. The data proves that deep discounts cannot compensate for product failure.



Pricing Strategy

Stop aggressive discounting on low-rated items. It attracts customers but results in negative reviews. Focus discounts on 4.0+ rated items to drive volume.



Inventory Mix

Bundle high-volume accessories (HDMI cables, memory cards) with high-margin electronics (TVs, Cameras) to increase basket size.

Analysis based on 1,350+ products and ~27M reviews.